Math Kangaroo Practice

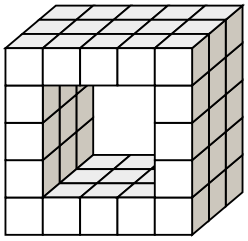
Grade 3-4 · 40 questions · harder · seed 41617

Part A: Each question is worth 3 points

1. The digits from 1 to 9 are written on 9 cards. Fred has 1, 6 and 7; Leo has 4, 5 and 9; Nina has 2, 3 and 8. Each of them uses some of +, –, ×, ÷ (and brackets) and each of their own cards exactly once.
Who **cannot** get 41 as the result?

A. Fred ***B.*** Leo ***C.*** Nina
D. All can get 41 ***E.*** Leo and Nina

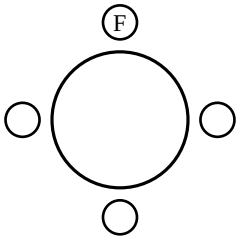
2. Robert built this tunnel out of identical cubes. The hole goes straight through from the front to the back.



How many cubes did Robert use?

A. 16 ***B.*** 48 ***C.*** 60 ***D.*** 64 ***E.*** 75

3. Four friends, Felix, Ben, Iris and Clara, go to a restaurant and sit at a table with 4 chairs. Felix always sits on the same chair (marked F).



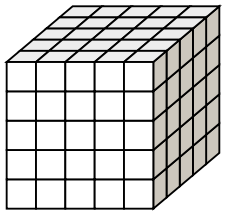
In how many different ways can the friends sit around the table?

A. 3 ***B.*** 6 ***C.*** 9 ***D.*** 24 ***E.*** 25

4. Mia, who is older than Sara by 3 years and 2 days, was born on January 6, 2014.
When was Sara born?

A. January 9, 2017 ***B.*** January 8, 2018 ***C.*** January 4, 2011
D. January 8, 2017 ***E.*** January 6, 2017

5. A cube with sides of 5 cm is painted grey all over and then cut into small cubes with sides of 1 cm.



How many of the small cubes have exactly 2 painted faces?

A. 60 ***B.*** 54 ***C.*** 40 ***D.*** 39 ***E.*** 36

6. To write all the house numbers in a street, starting from 1, exactly 105 digits are needed. How many houses are there?

A. 105 ***B.*** 66 ***C.*** 57 ***D.*** 49 ***E.*** 48

Part B: Each question is worth 4 points

7. What is the smallest number whose digits add up to 24?

A. 605 ***B.*** 662 ***C.*** 699 ***D.*** 700 ***E.*** 996

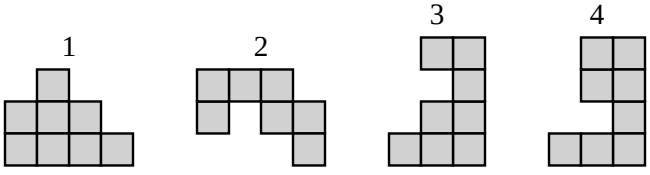
8. Aisha, Emma, Sofia, Omar and Leo were born in different months: December, October, April, January and August. Aisha has a birthday in summer. Omar was born in spring. Leo was born in autumn. Emma was born in a month that doesn't start with J.
Which statement is true?

A. Emma was born in April ***B.*** Aisha was born in December ***C.*** Leo was born in October
D. Leo was born in December ***E.*** Omar was born in October

9. Which word has exactly 2 different letters in common with SPIDER?

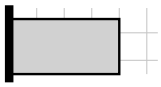
A. BANANA ***B.*** KANGAROO ***C.*** CHOCOLATE ***D.*** PENCIL ***E.*** TEACHER

10. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



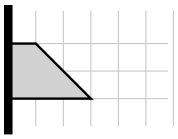
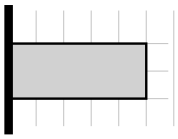
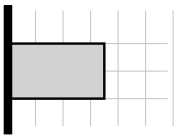
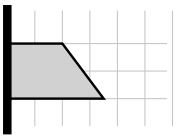
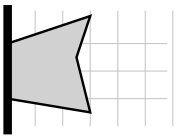
A. 1 and 2 ***B.*** 3 and 4 ***C.*** 1 and 4 ***D.*** 2 and 4 ***E.*** 1 and 3

11. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Nina draws the shadow at five different moments. Spread out flat, the flag is 4 squares wide and 2 squares tall (see the picture).



the flag, flat

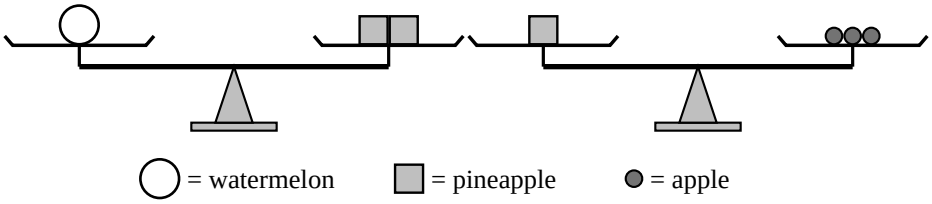
Which shadow **cannot** be right?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

12. A counter shows 53336. Every second it goes up by 1.
How many seconds until it shows a number whose 5 digits are all the same?

A. 13330 ***B.*** 8892 ***C.*** 2475 ***D.*** 2220 ***E.*** 2219

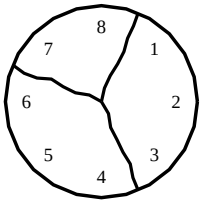
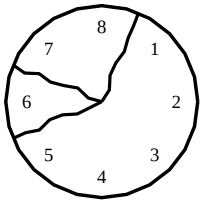
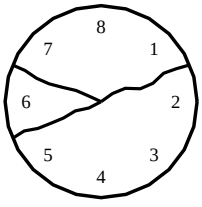
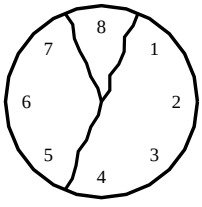
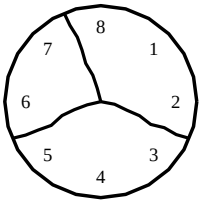
13. The two balances are level. One watermelon weighs as much as 2 pineapples, and one pineapple weighs as much as 3 apples.



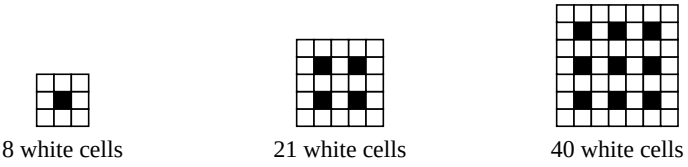
How many apples weigh as much as 3 watermelons?

A. 5 ***B.*** 6 ***C.*** 7 ***D.*** 15 ***E.*** 18

14. A wheel of fortune with the numbers 1 to 8 around its edge is cracked into 3 pieces. The sums of the numbers in the pieces are three consecutive numbers.
Which picture shows how it is cracked?

A.  ***B.***  ***C.***  ***D.***  ***E.*** 

15. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many white cells will the 7th diagram have?



A. 225 ***B.*** 196 ***C.*** 179 ***D.*** 176 ***E.*** 147

16. Only the even numbers from 2 to 200 are written down: 2, 4, 6, ..., 200.
How many digits are written?

A. 100 ***B.*** 245 ***C.*** 247 ***D.*** 249 ***E.*** 290

17. Leo, Tom, Lucas, Zoe and Sofia were born in different months: December, June, July, August and March. Zoe was born in spring. Leo was born in a month that starts with J. Tom was born in winter. Sofia was not born in July. Sofia was born in a month that starts with J.
Which statement is true?

A. Leo was born in July ***B.*** Lucas was born in July ***C.*** Zoe was born in July
D. Zoe was born in August ***E.*** Leo was born in August

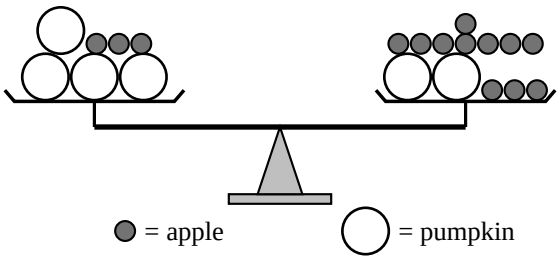
18. Dan's snake toy is 21 dm 1 cm long. Eva's snake toy is 1570 mm long.
How many centimetres longer is Dan's snake?

A. 54 cm ***B.*** 55 cm ***C.*** 64 cm ***D.*** 196 cm ***E.*** 540 cm

19. The odometer of a car shows 965628 km. Some of its digits are the same.
How many kilometres ago did the odometer last show a number with all different digits?

A. 142 ***B.*** 141 ***C.*** 140 ***D.*** 131 ***E.*** 1

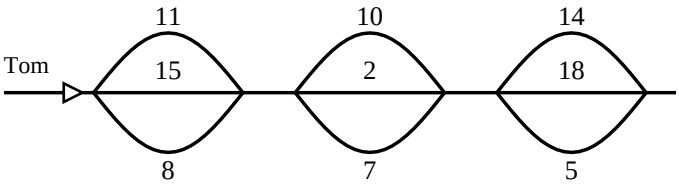
20. The balance is level: 4 pumpkins and 3 apples on one plate, 2 pumpkins and 11 apples on the other.



One pumpkin weighs the same as how many apples?

A. 4 apples ***B.*** 5 apples ***C.*** 8 apples ***D.*** 11 apples ***E.*** 14 apples

21. Tom walks along the road from left to right and adds up the numbers along the way.



Which total can Tom **not** get?

A. 15 ***B.*** 16 ***C.*** 30 ***D.*** 33 ***E.*** 39

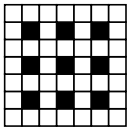
22. Nina thinks of a number, multiplies by 3, then takes away 1, then divides by 4, and gets 5. Which number did Nina think of?

A. 9 ***B.*** 8 ***C.*** 7 ***D.*** 6 ***E.*** 5

23. The diagrams below are the 1st, 2nd and 3rd of a pattern. How many cells in all will the 6th diagram have?

8 white cells

21 white cells


40 white cells

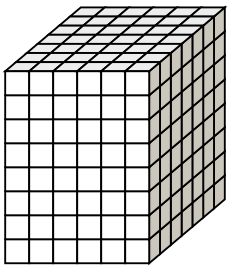
A. 36 ***B.*** 52 ***C.*** 133 ***D.*** 144 ***E.*** 169

Part C: Each question is worth 5 points

24. Exactly 540 digits are used to write all the page numbers in a book, starting from 1. How many pages are there?

A. 216 ***B.*** 215 ***C.*** 180 ***D.*** 126 ***E.*** 117

25. Chris builds a brick 6 cubes long, 7 cubes deep and 8 cubes tall from red and blue cubes. All the cubes you can see from the outside are red. The cubes just below them are **also** red, so on every side (top, bottom, front, back, left and right) there are two layers of red cubes. All the other cubes, in the middle, are blue.



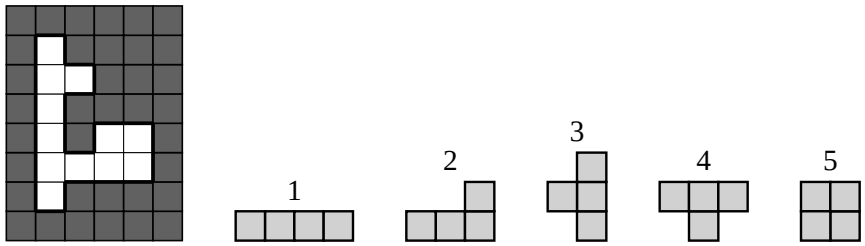
How many blue cubes are there?

A. 312 ***B.*** 120 ***C.*** 48 ***D.*** 24 ***E.*** 6

26. A big cube made of small cubes is painted on the outside and taken apart. Exactly 24 small cubes have 2 painted faces.
How many small cubes long is each edge of the big cube?

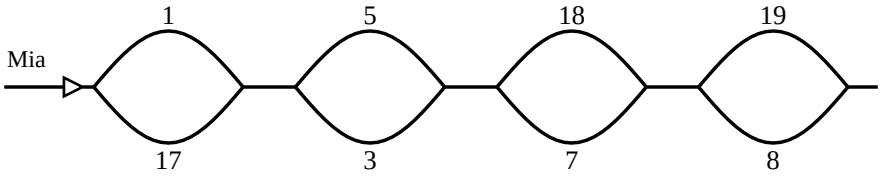
A. 7 ***B.*** 6 ***C.*** 5 ***D.*** 4 ***E.*** 3

27. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+4 ***C.*** 2+3+4 ***D.*** 2+4+5 ***E.*** 3+4+5

28. Mia walks along the road from left to right and adds up the numbers along the way. The total is 21.



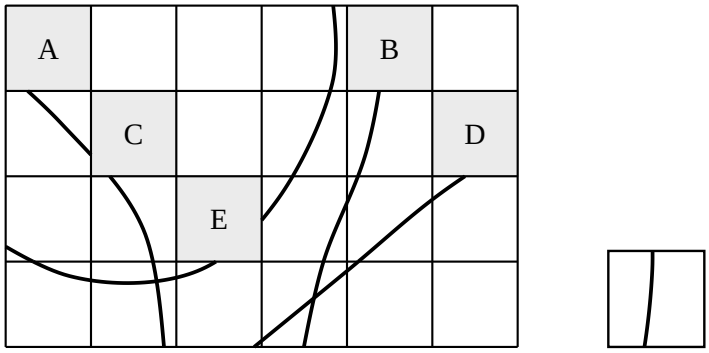
Which number did Mia see at the second fork?

A. 1 ***B.*** 3 ***C.*** 5 ***D.*** 8 ***E.*** 18

29. A bar code is made of 20 black and white bars taking turns. It starts with a black bar and ends with a **white** bar. Each black bar is wide or narrow. There are 1 more white bars than wide black bars.
How many narrow black bars are there?

A. 9 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

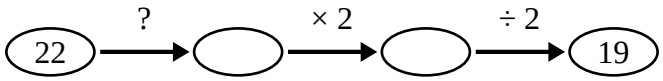
30. A cat played with balls of wool on squared paper. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

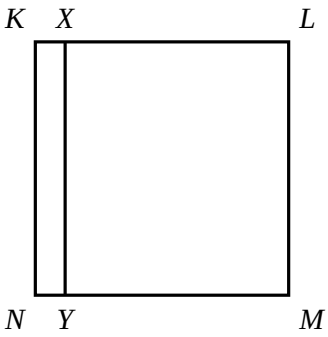
31. The arrows show calculations done one after the other, but one arrow has lost its calculation: it shows a question mark.



Which calculation belongs on that arrow?

A. + 1 ***B.*** × 3 ***C.*** + 7 ***D.*** + 3 ***E.*** – 3

32. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 38 cm and the perimeter of *XLMY* is 64 cm.



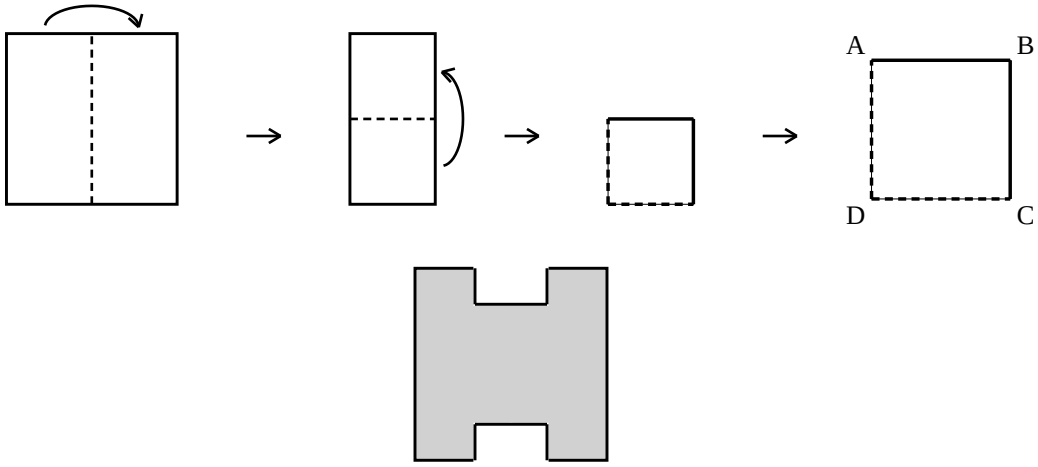
What is the perimeter of the square *KLMN*?

A. 102 cm ***B.*** 85 cm ***C.*** 84 cm ***D.*** 73 cm ***E.*** 68 cm

33. Finn's new rug is 3 m 7 cm long. Four of these lengths are the same as that. Which one is **not** the same?

A. 3 m 70 mm ***B.*** 30 dm 7 cm ***C.*** 3070 mm ***D.*** 37 cm ***E.*** 307 cm

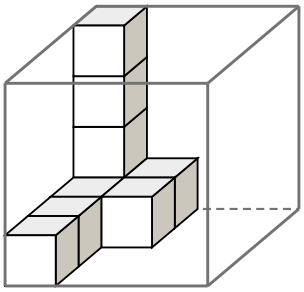
34. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square was cut off, and after unfolding the paper looks like the bottom picture.



Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

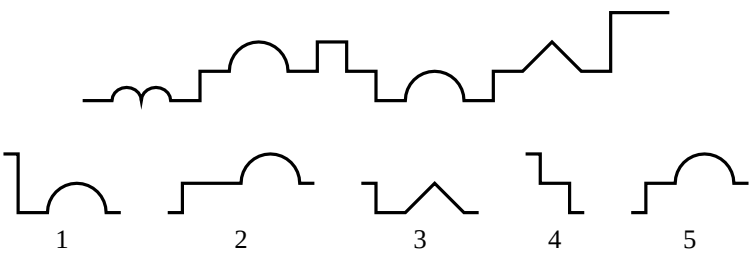
35. Tom puts cubes into a glass box, as in the picture. Every cube stands on the bottom or on another cube. Tom wants the two bottom layers to be completely full.



How many more cubes are needed for that?

A. 25 ***B.*** 24 ***C.*** 23 ***D.*** 19 ***E.*** 9

36. This is the outline of a town seen from the river at sunset.



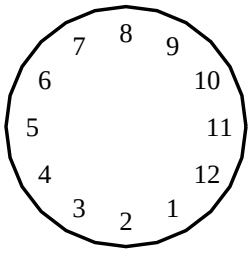
How many of the pieces 1–5 below belong to the line?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

37. Leo types the numbers from 1 to 150 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,150.
How many keys does Leo press (each digit and each comma is one key)?

A. 500 ***B.*** 493 ***C.*** 492 ***D.*** 491 ***E.*** 490

38. A round cake with the numbers 1 to 12 around its edge is broken by one straight crack into two pieces. The sums of the numbers on the two pieces are equal.



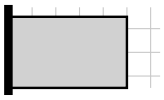
Where does the crack go?

A. between 10 and 11, and between 5 and 6
B. between 9 and 10, and between 3 and 4
C. between 7 and 8, and between 5 and 6
D. between 11 and 12, and between 4 and 5
E. between 8 and 9, and between 4 and 5

39. How many three-digit numbers have digits that add up to 7?

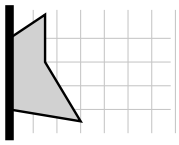
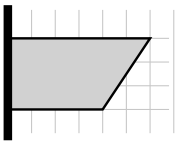
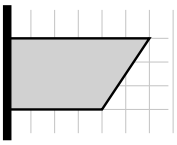
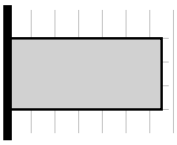
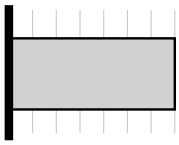
A. 56 ***B.*** 33 ***C.*** 32 ***D.*** 29 ***E.*** 28

40. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Tom draws the shadow at five different moments. Flat, the flag is 5 squares wide and 3 squares tall (see the picture). Four of the drawings are wrong.



the flag, flat

Which drawing can be right?

A. 
B. 
C. 
D. 
E. 

Answer Key

1.	**C**	**2.**	**B**	**3.**	**B**	**4.**	**D**	**5.**	**E**	**6.**	**C**
7.	**C**	**8.**	**C**	**9.**	**E**	**10.**	**E**	**11.**	**B**	**12.**	**E**
13.	**E**	**14.**	**E**	**15.**	**D**	**16.**	**C**	**17.**	**A**	**18.**	**A**
19.	**B**	**20.**	**A**	**21.**	**B**	**22.**	**C**	**23.**	**E**	**24.**	**A**
25.	**D**	**26.**	**D**	**27.**	**E**	**28.**	**C**	**29.**	**D**	**30.**	**B**
31.	**E**	**32.**	**E**	**33.**	**D**	**34.**	**A**	**35.**	**A**	**36.**	**B**
37.	**D**	**38.**	**B**	**39.**	**E**	**40.**	**A**				

Solutions

1. C (Nina) — Fred: 6 × 7 – 1 = 41; Leo: 5 × 9 – 4 = 41. Nina: with 2, 3 and 8 nothing gives 41 (the closest are 40 = (2 + 3) × 8 and 48 = 2 × 3 × 8). [2002 Q21 · who_cannot]

2. B (48) — The front has 5 × 5 – 3 × 3 = 16 cubes. The tunnel is 3 cubes long, so 16 × 3 = 48. [2002 Q20 · count_tunnel]

3. B (6) — Felix's chair is fixed. The other 3 friends fill the other chairs: 3 × 2 × 1 = 6 ways. [2002 Q22 · fixed_seat]

4. D (January 8, 2017) — Older means born earlier, so Sara was born 3 years and 2 days **after** January 6, 2014: 3 years later is January 6, 2017, then 2 more days: January 8, 2017. [2007 Q12 · younger_birthday]

5. E (36) — They sit on the edges, but not at the corners. 12 edges with 5 – 2 = 3 such cubes each: 12 × 3 = 36. [2007 Q17 · two_faces]

6. C (57) — 1 to 9 use 9 digits. The other 105 – 9 = 96 digits belong to two-digit numbers: 96 ÷ 2 = 48 numbers, from 10 to 57. So there are 57. [2003 Q16 · pages_from_digits]

7. C (699) — Use as few digits as possible: as many 9s as possible, with the smallest digit in front. 699: 6 + 9 + 9 = 24. [2007 Q4 · smallest_with_sum]

8. C (Leo was born in October) — The only possible matching: Aisha – August; Emma – December; Sofia – January; Omar – April; Leo – October. [2002 Q18 · which_true]

9. E (TEACHER) — TEACHER shares E and R with SPIDER. BANANA shares 0; PENCIL shares 3; KANGAROO shares 1; CHOCOLATE shares 1. [2007 Q3 · which_word]

10. E (1 and 3) — Pieces 1 and 3 have 16 squares together and fit into a 4 × 4 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

11. B — Turning or folding can only make the flag look narrower. Picture B reaches more than 4 squares away from the pole, which is wider than the flag. [2002 Q15 · too_wide]

12. E (2219) — The numbers with all digits the same are 44444, 55555, ... The next one after 53336 is 55555: 55555 – 53336 = 2219. [2002 Q24 · same_digits]

13. E (18) — One watermelon = 2 pineapples = 2 × 3 = 6 apples; 3 watermelons = 3 × 6 = 18 apples. [2002 Q9 · chain]

14. E — All numbers add up to 36, so the pieces must be 11 + 12 + 13 = 36. Only this picture gives 11, 12, 13. [2002 Q11 · which_picture]

15. D (176) — Diagram number k is (2k + 1) × (2k + 1) with k × k black cells. The 7th: 15 × 15 – 7 × 7 = 225 – 49 = 176. [2007 Q21 · far_diagram]

16. C (247) — Even one-digit numbers: 4; even two-digit numbers: 45; even three-digit numbers: 51. 4 × 1 + 45 × 2 + 51 × 3 = 247. [2007 Q23 · even_only]

17. A (Leo was born in July) — The only possible matching: Leo – July; Tom – December; Lucas – August; Zoe – March; Sofia – June. [2002 Q18 · which_true]

18. A (54 cm) — 21 dm 1 cm = 211 cm and 1570 mm = 157 cm. 211 – 157 = 54 cm. [2003 Q10 · how_much_longer]

19. B (141) — Counting down from 965628, the first number with all different digits is 965487: 965628 – 965487 = 141. [2002 Q24 · last_time]

20. A (4 apples) — Take 2 pumpkins and 3 apples off both plates. Left: 2 pumpkins = 8 apples, so one pumpkin = 8 ÷ 2 = 4. [2002 Q9 · both_sides]

21. B (16) — 15 = 8 + 2 + 5; 30 = 15 + 10 + 5; 33 = 8 + 7 + 18; 39 = 11 + 10 + 18. No route gives 16. [2007 Q1 · impossible_total]

22. C (7) — Go backwards: 5 × 4 = 20, 20 + 1 = 21, 21 ÷ 3 = 7. [2007 Q5 · think_of_a_number]

23. E (169) — The 6th diagram is 13 × 13 = 169 cells, with 6 × 6 = 36 black ones. [2007 Q21 · black_and_total]

24. A (216) — 1-9 use 9 digits, 10-99 use 90 × 2 = 180, together 189. The other 540 – 189 = 351 digits are three-digit numbers: 351 ÷ 3 = 117, from 100 to 216. [2003 Q16 · three_digit_pages]

25. D (24) — Each length loses two red cubes at each end, so it drops by 4: 2 × 3 × 4 = 24. [2003 Q9 · thick_wall]

26. D (4) — A cube has 12 edges, so each edge has 24 ÷ 12 = 2 cubes between its corners. With the 2 corners, an edge is 4 cubes long. [2007 Q17 · how_big]

27. E (3+4+5) — The empty area has 12 squares, so the three sizes must add up to 12. Only 3 + 4 + 5 also fit together in the right shape. [2003 Q18 · three_pieces]

28. C (5) — Only one route gives 21: 1 + 5 + 7 + 8 = 21. At the second fork: 5. [2007 Q1 · which_branch]

29. D (1) — Starting black and ending white, the bars come in pairs: 10 black and 10 white. Wide: 10 – 1 = 9. Narrow: 10 – 9 = 1. [2003 Q17 · ends_differ]

30. B (gap B) — The square has top: 1 line, right: no line, bottom: 1 line, left: no line. Only gap B has lines reaching it at exactly those places. [2002 Q1 · which_gap]

31. E (– 3) — Before that arrow we have 22; working back from 19, after it we need 19. Only – 3 turns 22 into 19. [2007 Q5 · which_operation]

32. E (68 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 38 + 64 = 102 cm = 6 sides. Side = 102 ÷ 6 = 17 cm, perimeter 4 × 17 = 68 cm. [2002 Q6 · two_perimeters]

33. D (37 cm) — 3 m 7 cm = 307 cm = 3070 mm. But 37 cm = 37 cm. [2003 Q10 · not_equal]

34. A (corner A) — The design shows notches in the middle of the top and bottom sides. That comes from cutting corner A (where one fold meets a paper edge). [2007 Q24 · which_corner]

35. A (25) — Two layers hold 2 × 4 × 4 = 32 cubes. In those two layers there are already 7 cubes (columns count at most 2 each). 32 – 7 = 25. [2007 Q14 · two_layers]

36. B (2) — Pieces 4, 5 fit. Not: 1 (its step is taller than in the line); 2 (its flat part between the two features has the wrong length); 3 (it steps down where the line steps up). [2002 Q7 · how_many_fit]

37. D (491) — Digits: 9 × 1 + 90 × 2 + 51 × 3 = 342. Commas: one fewer than the numbers, 149. 342 + 149 = 491. [2007 Q23 · with_commas]

38. B (between 9 and 10, and between 3 and 4) — Total 78, so each piece has 39: 10+11+12+1+2+3 = 39 and 4+5+6+7+8+9 = 39. [2002 Q11 · straight_crack]

39. E (28) — Count by the first digit: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28. [2007 Q4 · count_with_sum]

40. A — Check two things: the pole edge must be covered all the way, and nothing may reach more than 5 squares from the pole. Only picture A passes both checks. [2002 Q15 · only_one_right]